

Derivation of Sterile Neutrinos in the E_6 Inspired Model

In the E_6 inspired supersymmetric model described in [arXiv:1504.00904](https://arxiv.org/abs/1504.00904), the neutral sector for each generation is expanded beyond the Standard Model (SM) to include several singlets and vector-like states.

1. Neutral Sector Basis

The matter content of each generation is contained in the **27** representation of E_6 . Under the $SO(10) \times U(1)_\psi$ subgroup, the **27** decomposes as:

$$27 = 16(1) + 10(-2) + 1(4)$$

The neutral states in each generation are:

- ν : The light neutrino (part of the L doublet in the **16**).
- ν^c : The right-handed neutrino (SM singlet in the **16**).
- s : The $SO(10)$ singlet (SM singlet in the **1**).
- ν' and ν'^c : Components of the vector-like lepton doublets L_0 and L_0^c in the **10**.

For a two-family benchmark, the basis is:

$$\Psi_N = (\nu_1, \nu_2, \nu'_1, \nu'_2, \nu_1^c, \nu_2^c, s_1, s_2, \nu_1'^c, \nu_2'^c)^T$$

2. Mass Matrix Structure

The 10×10 neutral mass matrix M_N has the following block structure (suppressing generation indices):

$$\mathcal{L}_{mass} = \frac{1}{2} \Psi_N^T M_N \Psi_N + h.c.$$

The matrix M_N is derived from the Yukawa couplings Y_{27} and $Y_{351'}$ and the various VEVs. Based on the paper, the mass terms involve:

- **Dirac terms** mixing ν, ν' with ν^c, s, ν'^c .
- **Majorana terms** for ν^c and s induced by the E_6 breaking VEVs (e.g., f_1 and f_3).

3. The Sterile Neutrino Definition

In this context, the "sterile neutrinos" are the states that are singlets under the SM gauge group:

1. **Right-handed neutrinos (ν^c):** Part of the $SO(10)$ 16-plet.
2. E_6 **singlets (s):** The $SO(10)$ singlet state.

These states are typically very heavy (at the seesaw scale $\sim 10^{11} - 10^{16}$ GeV), which explains why the "sterile fractions" in the light neutrinos are so small ($\sim 10^{-20}$).

4. Seesaw and Mixing

The light neutrino mass matrix m_ν is obtained by the seesaw formula:

$$m_\nu \approx -m_D M_R^{-1} m_D^T$$

where m_D is the Dirac mass matrix and M_R is the heavy Majorana matrix.

The **sterile amplitude** in the light neutrino ν_{light} is given by the mixing element:

$$A_{sterile} \approx m_D M_R^{-1}$$

The **sterile fraction** is the square of this amplitude:

$$P_{sterile} \approx |A_{sterile}|^2$$

Given your Point 1 results:

- $m_{\nu^c} \sim 10^{11}$ GeV
- $m_s \sim 10^{15}$ GeV
- Sterile amplitudes $\sim 10^{-10}$
- Sterile fractions $\sim 10^{-20}$

This confirms that the sterile states are decoupled and the light neutrinos are almost purely active.